

Pathobiology of obesity and osteoarthritis: integrating biomechanics and inflammation



Obesity is a significant risk factor for developing osteoarthritis in weight-bearing and non-weight-bearing joints. Although the pathogenesis of obesity-associated osteoarthritis is not completely understood, recent studies indicate that pro-inflammatory metabolic factors contribute to an increase in osteoarthritis risk. Adipose tissue, and in particular infrapatellar fat, is a local source of pro-inflammatory mediators that are increased with obesity and have been shown to increase cartilage degradation in cell and tissue culture models. One adipokine in particular, leptin, may be a critical mediator of obesity-associated osteoarthritis via synergistic actions with other inflammatory cytokines. Biomechanical factors may also increase the risk of osteoarthritis by activating cellular inflammation and promoting oxidative stress. However, some types of biomechanical stimulation, such as physiologic cyclic loading, inhibit inflammation and protect against cartilage degradation. A high percentage of obese individuals with knee osteoarthritis are sedentary, suggesting that a lack of physical activity may increase the susceptibility to inflammation. A more comprehensive approach to understanding how obesity alters daily biomechanical exposures within joint tissues may provide new insight into the protective and damaging effects of biomechanical factors on inflammation in osteoarthritis.